

Scalability & Future Plan

Date	15 March 2024
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Supports a limited number of concurrent users on a local server.	Reduced performance under heavy load.	High
2	Uses SQLite database, which is suitable for small-scale applications.	Limited scalability for large datasets and multiple users	Medium
3	Uses SQLite database, which is suitable for small-scale applications. Reduced performance under heavy load.	Limited analysis and fewer prediction insights.	Medium Supports local users only. Supports local users only.

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports local users only.	Deploy on cloud platforms (Render, AWS, Azure) with load balancing.
2	Data Storage	SQLite database.	Migrate to MySQL or PostgreSQL for better scalability and performance.
3	Performance	Single Flask application.	Optimize model inference, implement caching, and use Gunicorn/Nginx for production deployment.
4	Security	Basic user authentication.	Add JWT authentication, HTTPS, role-based access control, and encrypted password storage

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Country comparison dashboard, interactive charts, and advanced analytics.	Next 3 Months	Better visualization and comparative analysis of HDI across countries.
Phase 3	Real-time data integration using World Bank and UNDP APIs.	Next 6 Months	More accurate and up-to-date HDI predictions using live data.
Phase 4	AI-powered policy recommendations, cloud deployment, and mobile application support.	Next 12 Months	Increased accessibility, scalability, and decision-support capabilities for users.